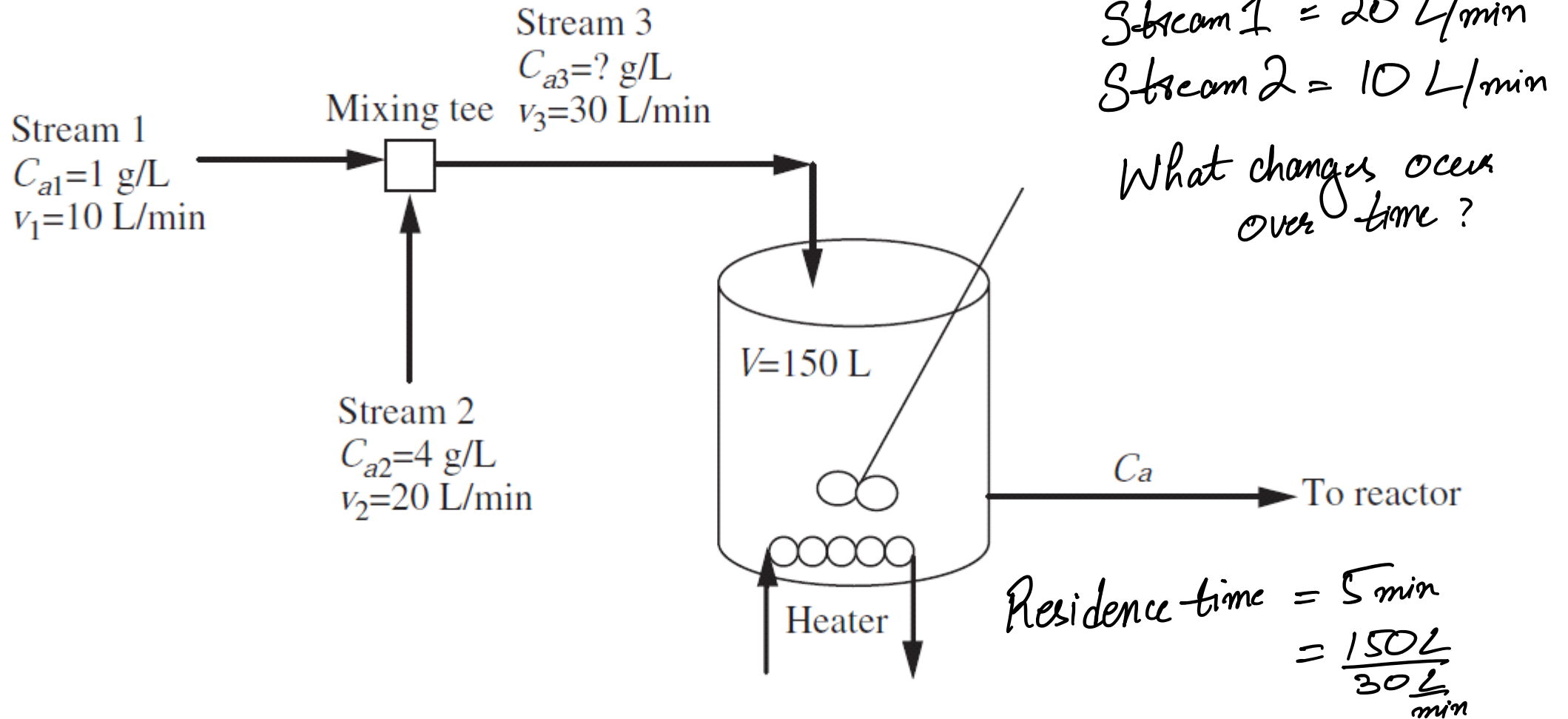


Modeling Tools for Process Dynamics

Chemical Mixing & Heating- Process Dynamics



Mining tee $\rightarrow$ Mass Balance

For Component A

Input from Stream 1 + Input from Stream 2 = Output in Stream 3

$$v_1 C_{a1} + v_2 C_{a2} = v_3 C_{a3}$$

flow rate $\downarrow$ conc. $\downarrow$

Before
Disturbance

$$\left(10 \frac{\text{L}}{\text{min}} \times 1 \frac{\text{g}}{\text{L}}\right) + \left(20 \frac{\text{L}}{\text{min}} \times 4 \frac{\text{g}}{\text{L}}\right) = 30 \frac{\text{L}}{\text{min}} \times C_{a3} \frac{\text{g}}{\text{L}}$$

$$\left(10 \frac{\text{g}}{\text{min}} + 80 \frac{\text{g}}{\text{min}}\right) \times \frac{1}{30} \frac{\text{min}}{\text{L}} = C_{a3} \frac{\text{g}}{\text{L}} = 3 \frac{\text{g}}{\text{L}}$$

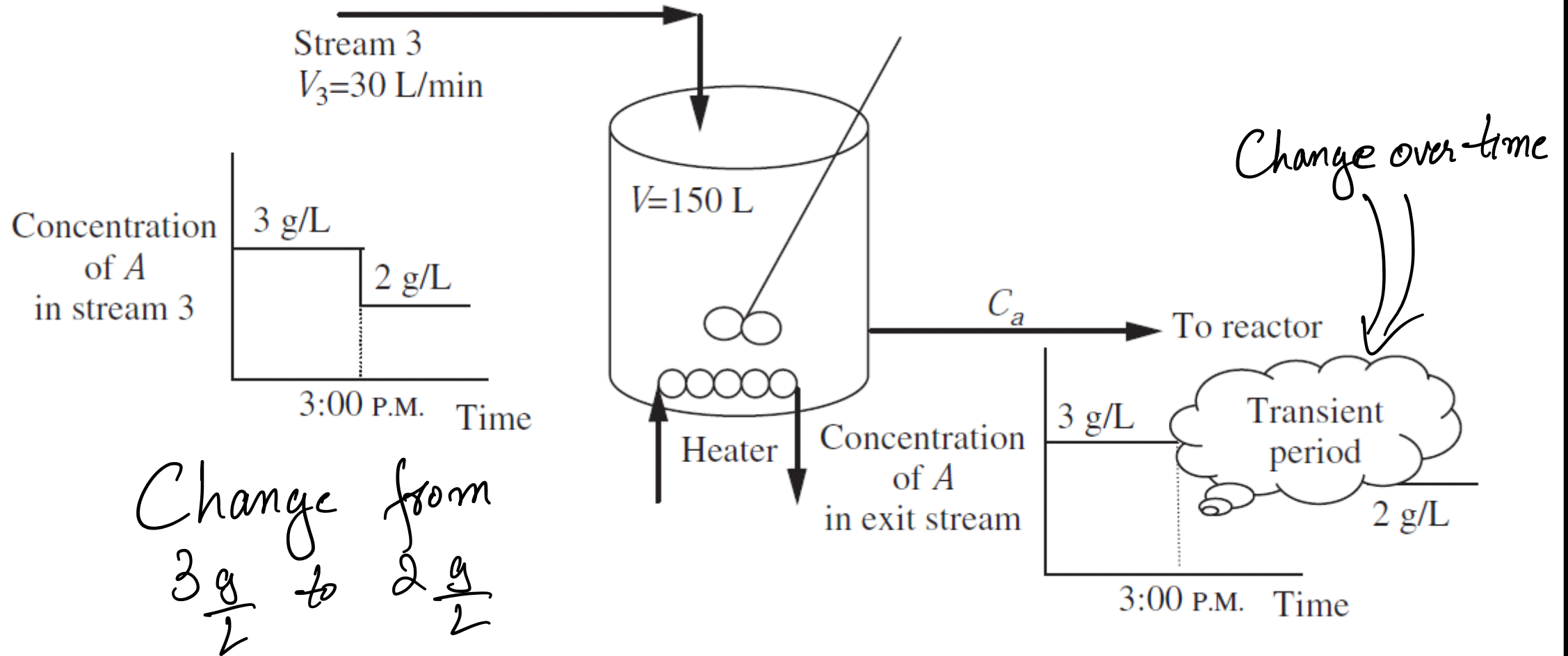
After
Disturbance

$$\left(20 \frac{\text{L}}{\text{min}} \times 1 \frac{\text{g}}{\text{L}}\right) + \left(10 \frac{\text{L}}{\text{min}} \times 4 \frac{\text{g}}{\text{L}}\right) = 30 \frac{\text{L}}{\text{min}} \times C_{a3} \frac{\text{g}}{\text{L}}$$

$$\Rightarrow C_{a3} = 2 \frac{\text{g}}{\text{L}}$$

Decrease

Change in Exit Con. of Heating Vessel



Analysis of Exit variation with time

→ Unsteady Mass Balance on component A

Input of A - Output of A = Accumulation of A

$$v_3 C_{a3} - v_3 C_a = \frac{d}{dt} (V C_a)$$

$$v_3 (C_{a3} - C_a) = \frac{d}{dt} (V C_a)$$

$$C_{a3} - C_a = \underbrace{\frac{V}{v_3}}_{\tau} \frac{dC_a}{dt}$$

$\tau \rightarrow$ Residence time = 5 min

$$\tau \frac{dC_a}{dt} + C_a - C_{a3} = 0$$

$$@ t=0 ; C_a(0) = 3 \frac{g}{L}$$

After Disturbance

$$C_{a3} = 2 \frac{g}{L}$$

$$\Rightarrow 5 \frac{dC_a}{dt} + C_a = 2$$

$$\int_3^{C_a} \frac{dC_a}{2-C_a} = \int_0^t \frac{dt}{5}$$

$$-\ln(2-C_a) \Big|_3^{C_a} = \frac{t}{5} \Big|_0^t$$

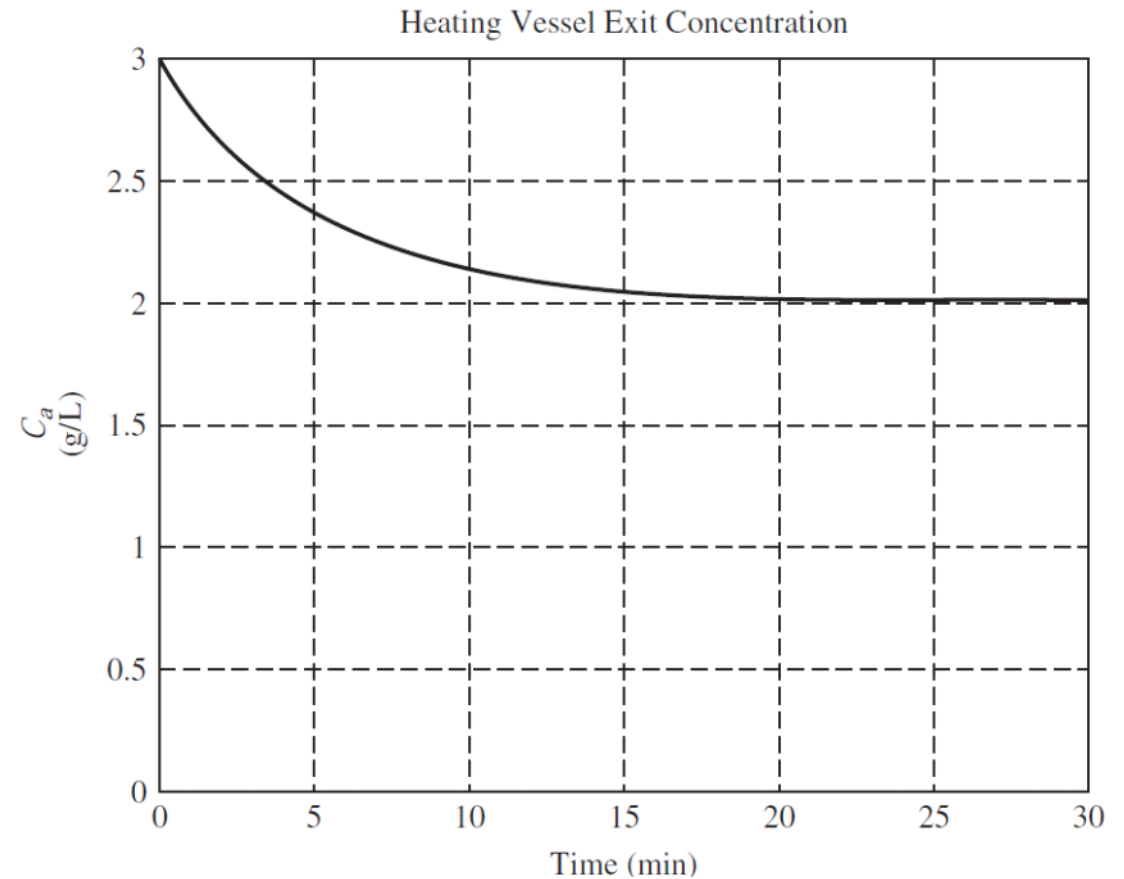
$$-\ln(2-C_a) - (-\ln(2-3)) = \frac{t}{5}$$

$$\ln\left(\frac{2-C_a}{2-3}\right) = -\frac{t}{5}$$

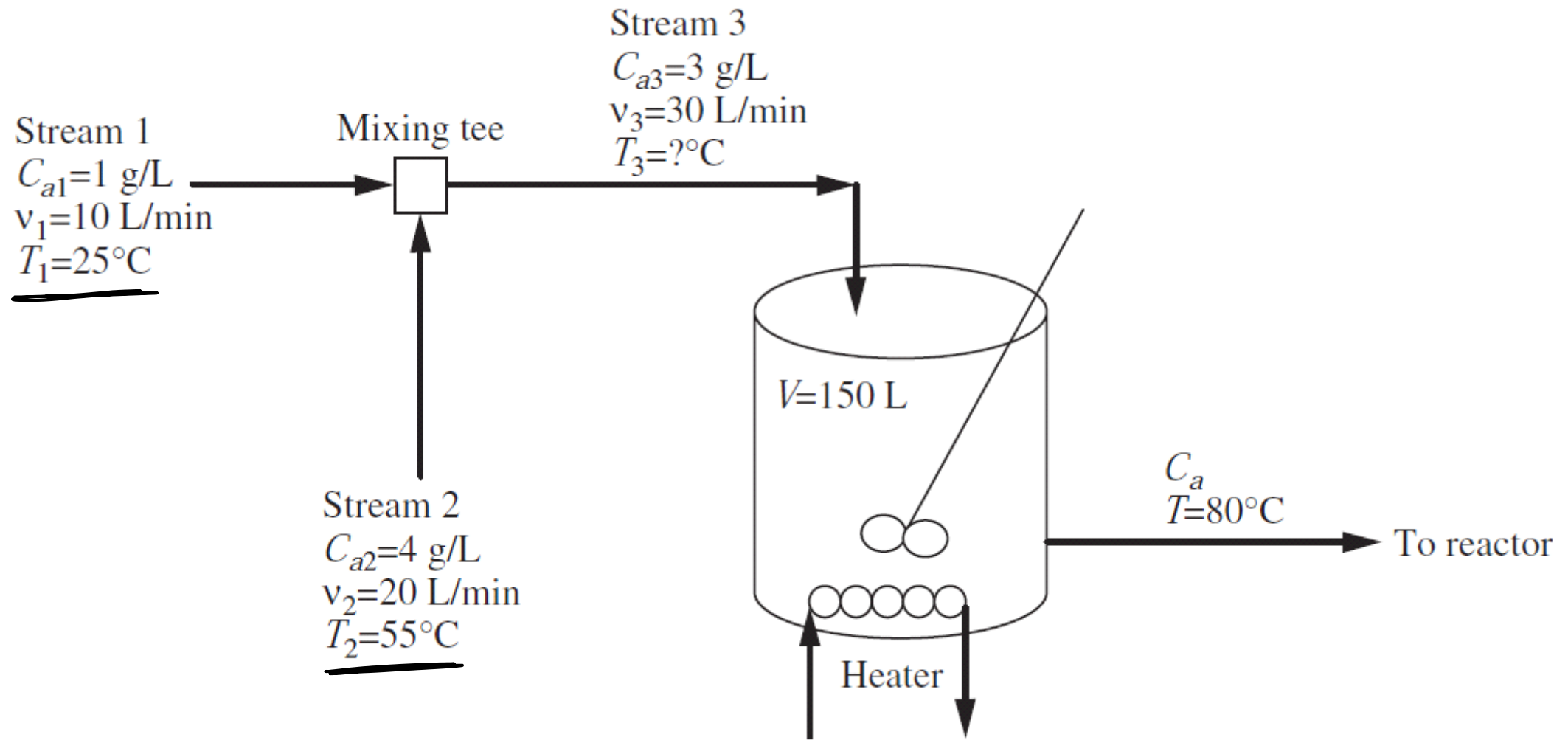
$$\frac{2-C_a}{2-3} = e^{-\frac{t}{5}}$$

$$2-C_a = -e^{-t/5}$$

$$\boxed{C_a = 2 + e^{-t/5}}$$



Energy Balance over the Process



Steady State Analysis for Mixing tee Before Disturbance

$$\begin{array}{c} \text{Input Enthalpy} \\ \text{for Stream 1} \end{array} + \begin{array}{c} \text{Input Enthalpy} \\ \text{for Stream 2} \end{array} = \begin{array}{c} \text{Output Enthalpy} \\ \text{for Stream 3} \end{array}$$

Using $H = m C_p \Delta T = \int V C_p \Delta T$

$$\int V_1 C_p (T_1 - T_{ref}) + \int V_2 C_p (T_2 - T_{ref}) = \int V_3 C_p (T_3 - T_{ref})$$

with constant $\int$ and C_p as λ come changes

$$V_1 (T_1 - T_{ref}) + V_2 (T_2 - T_{ref}) = V_3 (T_3 - T_{ref})$$

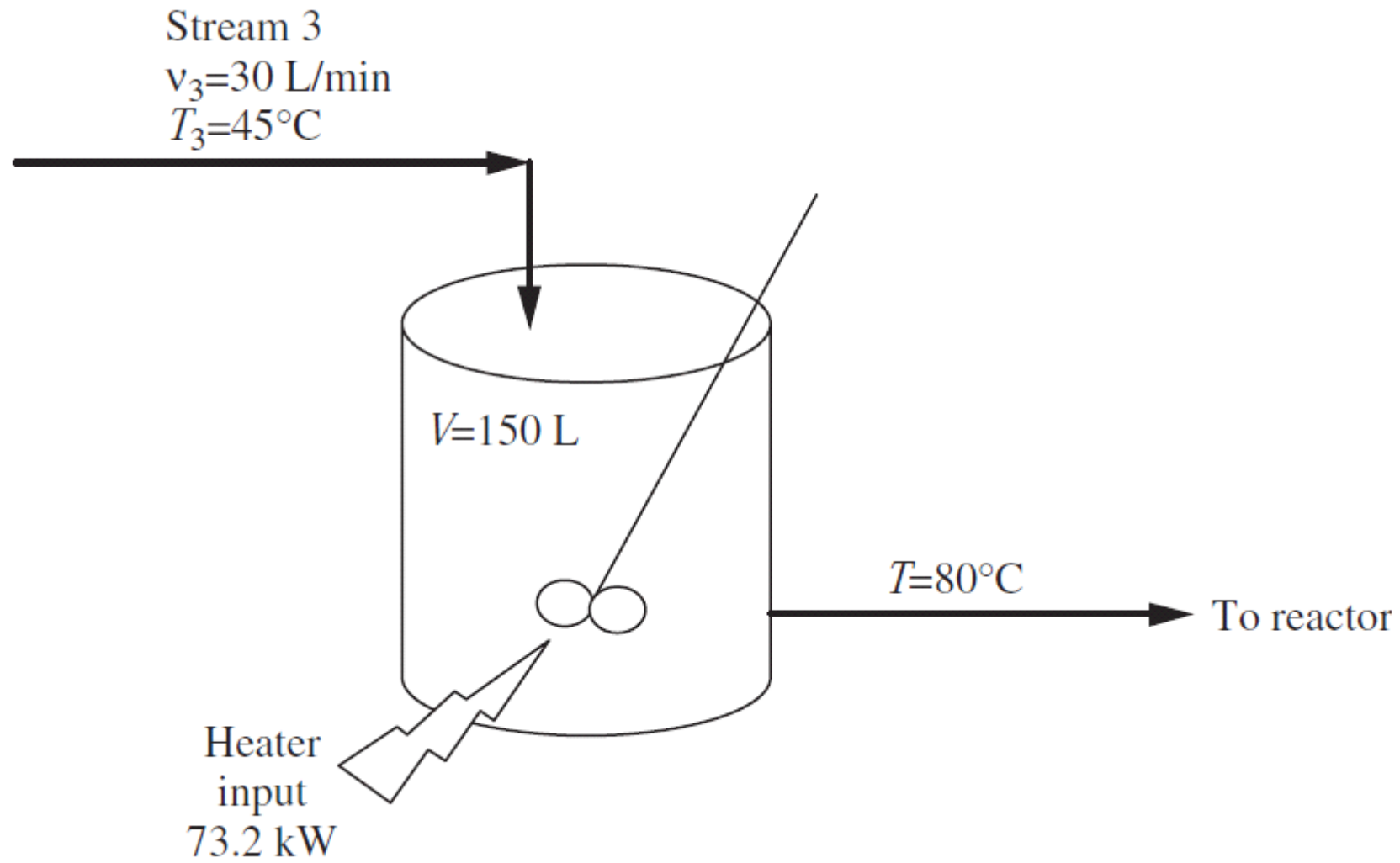
$$V_1 T_1 - V_1 T_{ref} + V_2 T_2 - V_2 T_{ref} = V_3 T_3 - V_3 T_{ref}$$

$$V_1 T_1 + V_2 T_2 - (V_1 + V_2) T_{ref} = V_3 T_3 - V_3 T_{ref}$$

$$\therefore (V_1 + V_2) T_{ref} = V_3 T_{ref}$$

$$\boxed{V_1 T_1 + V_2 T_2 = V_3 T_3}$$

Steady State $\Rightarrow T_3 = \frac{V_1 T_1 + V_2 T_2}{V_3} = \frac{(10)(25) + (20)(25)}{30} = 45^\circ\text{C}$



Steady State Analysis for Heating Vessel

Enthalpy input from stream 3 + Enthalpy input from heater = Enthalpy output of vessel

$$\dot{V}_3 C_p (T_3 - T_{ref}) + Q = \dot{V}_3 C_p (T - T_{ref})$$

$$Q = \dot{V}_3 C_p (T - T_3)$$

$$= \left(1000 \frac{\text{g}}{\text{L}}\right) \times 30 \frac{\text{L}}{\text{min}} \times 1 \frac{\text{cal}}{\text{g} \cdot ^\circ\text{C}} \times (80 - 45)^\circ\text{C} = 1.05 \times 10^6 \frac{\text{cal}}{\text{min}}$$

$$Q = 73.2 \text{ kW} \Rightarrow \text{Required}$$

After Disturbance

$$T_3 = \frac{v_1 T_1 + v_2 T_2}{v_3} = \frac{(20)(25) + (10)(55)}{30} = 35^\circ$$

Original S.S. $T = 45^\circ\text{C}$

Unsteady State Mass Balance

Enthalpy input from Stream 3 - Enthalpy output from heating vessel + Enthalpy input from heater = Enthalpy Accumulation in Vessel

$$\dot{V}_3 C_p (T_3 - T_{ref}) - \dot{V}_3 C_p (T - T_{ref}) + Q = \frac{d}{dt} (\rho V C_p (T - T_{ref}))$$

$$\dot{V}_3 C_p (T_3 - \cancel{T_{ref}} - T + \cancel{T_{ref}}) + Q = \rho V C_p \left(\frac{dT}{dt} - \frac{dT_{ref}}{dt} \right)$$

$$\dot{V}_3 C_p (T_3 - T) + Q = \rho V C_p \frac{dT}{dt}$$

$$T_3 - T + \frac{Q}{\dot{V}_3 C_p} = \underbrace{\frac{\rho V C_p}{\dot{V}_3 C_p}}_{\tau} \frac{dT}{dt}$$

$$\tau \frac{dT}{dt} + T = T_3 + \frac{Q}{\dot{V}_3 C_p}$$

$$\tau \frac{dT}{dt} + T = T_3 + \frac{Q}{F V_3 C_p}$$

$$(5 \text{ min}) \frac{dT}{dt} + T = 35^\circ\text{C} + \frac{(1.05 \times 10^6 \text{ cal/min})}{\left(1000 \frac{\text{g}}{\text{L}}\right) \left(30 \frac{\text{L}}{\text{min}}\right) \left(1 \frac{\text{cal}}{\text{g} \cdot ^\circ\text{C}}\right)} = 70^\circ\text{C}$$

$$5 \frac{dT}{dt} + T = 70$$

$$@ t=0 ; T = 80^\circ\text{C}$$

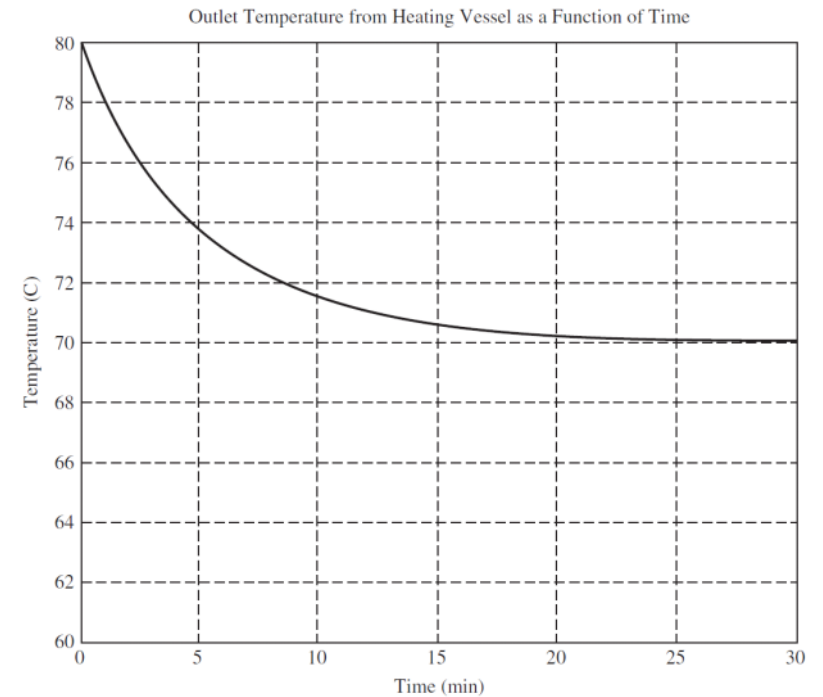
$$\int_{80}^T \frac{dT}{70-T} = \int_0^t \frac{dt}{5}$$

$$-\ln \left(\frac{70-T}{70-80} \right) = \frac{t}{5}$$

$$\frac{70-T}{-10} = e^{-\frac{t}{5}}$$

$$70-T = -10e^{-\frac{t}{5}}$$

$$T = 70 + 10e^{-\frac{t}{5}}$$



Mathematical Tools for Modeling

Laplace Transform

for function $f(t)$; $\mathcal{L}\{f(t)\} = \int_0^{\infty} f(t) e^{-st} dt = F(s)$

e.g for $f(t) = 1$

$$\begin{aligned} F(s) &= \mathcal{L}\{f(t)\} = \int_0^{\infty} 1 e^{-st} dt \\ &= \left. -\frac{1}{s} e^{-st} \right|_0^{\infty} \\ &= \frac{-e^{-s\infty}}{s} + \frac{e^{-s \times 0}}{s} \\ &= \frac{1}{s} = \mathcal{L}\{1\} \end{aligned}$$

Laplace Transform is Linear

$$\mathcal{L}\{a f_1(t) + b f_2(t)\} = a \mathcal{L}\{f_1(t)\} + b \mathcal{L}\{f_2(t)\}$$

Laplace Transforms of Simple functions

→ Step function

$$f(t) = \begin{cases} 0 & t < 0 \\ 1 & t > 0 \end{cases}$$

This is called unit step function $u(t)$

$$\mathcal{L}\{u(t)\} = \frac{1}{s} = F(s)$$

$$\text{Similarly } \mathcal{L}\{A u(t)\} = \frac{A}{s} = F(s)$$

→ Exponential function

$$f(t) = \begin{cases} 0 & t < 0 \\ e^{-at} & t > 0 \end{cases} = u(t)e^{-at}$$

$$\mathcal{L}\{u(t)e^{-at}\} = \int_0^{\infty} e^{-at} e^{-st} dt$$

$$= \int_0^{\infty} e^{-(a+s)t} dt$$

$$= \left[\frac{-1}{a+s} e^{-(a+s)t} \right]_0^{\infty}$$

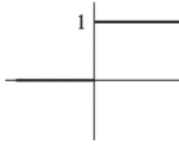
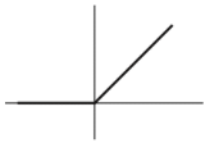
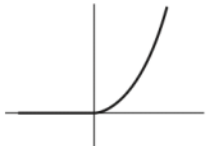
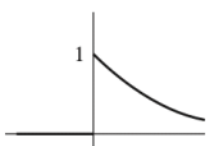
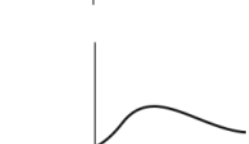
$$= \frac{-e^{-\infty}}{a+s} - \frac{-e^0}{a+s} = \frac{1}{a+s}$$

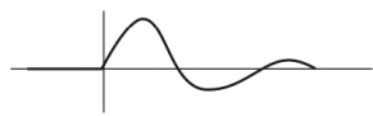
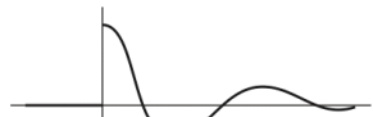
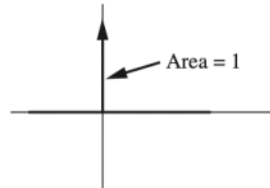
Ramp function

$$f(t) = \begin{cases} 0 & t < 0 \\ t & t > 0 \end{cases} = t u(t)$$

↪ Sine function

$$f(t) = \begin{cases} 0 & t < 0 \\ \sin kt & t > 0 \end{cases} = u(t) \sin kt$$

Function	Graph	Transform
$u(t)$		$\frac{1}{s}$
$tu(t)$		$\frac{1}{s^2}$
$t^n u(t)$		$\frac{n!}{s^{n+1}}$
$e^{-at} u(t)$		$\frac{1}{s + a}$
$t^n e^{-at} u(t)$		$\frac{n!}{(s + a)^{n+1}}$
$\sin kt u(t)$		$\frac{k}{s^2 + k^2}$

Function	Graph	Transform
$\cos kt u(t)$		$\frac{s}{s^2 + k^2}$
$\sinh kt u(t)$		$\frac{k}{s^2 - k^2}$
$\cosh kt u(t)$		$\frac{s}{s^2 - k^2}$
$e^{-at} \sin kt u(t)$		$\frac{k}{(s + a)^2 + k^2}$
$e^{-at} \cos kt u(t)$		$\frac{s + a}{(s + a)^2 + k^2}$
$\delta(t)$, unit impulse		1

Transforms of Derivatives

$$\mathcal{L}\left\{\frac{df(t)}{dt}\right\} = sF(s) - f(0) \quad ; \quad \text{where } F(s) = \mathcal{L}\{f(t)\}$$

Similarly for Second Derivatives

$$\begin{aligned}\mathcal{L}\left\{\frac{d^2f(t)}{dt^2}\right\} &= \mathcal{L}\left\{\frac{d}{dt}\left(\frac{df(t)}{dt}\right)\right\} = s \mathcal{L}\left\{\frac{df(t)}{dt}\right\} - \left.\frac{df(t)}{dt}\right|_{t=0} \\ &= s[sF(s) - f(0)] - f'(0) \\ &= s^2 F(s) - sf(0) - f'(0)\end{aligned}$$

Hence it follows that

$$\mathcal{L}\left\{\frac{d^n f(t)}{dt^n}\right\} = s^n F(s) - s^{n-1}f(0) - s^{n-2}f^{(1)}(0) - \dots - sf^{(n-2)}(0) - f^{(n-1)}(0)$$

Example

Laplace Transform of function $x(t)$ that satisfies differential eqn and initial conditions

$$\mathcal{L}\{f^n(t)\} = s^n F(s) - \sum_{k=0}^{n-1} s^{n-1-k} f^{(k)}(0)$$

$$\frac{d^3 x}{dt^3} + 4 \frac{d^2 x}{dt^2} + 5 \frac{dx}{dt} + 2x = 2$$

$$\text{and } x(0) = \frac{dx(0)}{dt} = \frac{d^2 x(0)}{dt^2} = 0$$

Taking Laplace of b.s of diff eqn

$$\left\{ \begin{aligned} &[s^3 x(s) - s^2 x(0) - s x'(0) - x''(0)] \\ &+ 4[s^2 x(s) - s x(0) - x'(0)] \\ &+ 5[s x(s) - x(0)] + 2x(s) \end{aligned} \right\} = \frac{2}{s}$$

$$s^3 x(s) + 4s^2 x(s) + 5s x(s) + 2x(s) = \frac{2}{s}$$

$$x(s) = \frac{2}{s(s^3 + 4s^2 + 5s + 2)}$$

Solution of Ordinary Differential Equations (ODEs)

→ O.D.Es converted to transforms

↳ Their resulting eqns are algebraically solved for unknown function

$$\frac{d^3 x}{dt^3} + 4 \frac{d^2 x}{dt^2} + 5 \frac{dx}{dt} + 2x = 2$$

$$\text{and } x(0) = \frac{dx(0)}{dt} = \frac{d^2 x(0)}{dt^2} = 0$$

⇒ Characteristic eqn
Particular solution
General solution
Initial conditions

Solution is

$$x(t) = 1 - 2te^{-t} - e^{-2t}$$

with Laplace transform

$$x(s) = \frac{1}{s} - \frac{2}{(s+1)^2} - \frac{1}{s+2} \Leftrightarrow$$

$$x(s) = \frac{2}{s(s^3 + 4s^2 + 5s + 2)}$$

Using These tools for our initial problem

$$\tau \frac{dC_a}{dt} + C_a = C_{a3} \quad - \text{Mass Balance}$$

$$\tau \frac{dT}{dt} + T = T_3 + \frac{Q}{FV_3 C_p} \quad - \text{Energy Balance}$$

Laplace Transform for Mass Balance

$$\tau [s C_a(s) - C_a(0)] + C_a(s) = C_{a3}(s)$$

$$(1 + \tau s) C_a(s) - \tau C_a(0) = C_{a3}(s)$$

$$C_a(s) = \frac{C_{a3}(s) + \tau C_a(0)}{\tau s + 1}$$

with $C_{a3} = \frac{2g}{L}$ after disturbance
 $C_a(0) = 3\frac{g}{L} \rightarrow$ initial conc. before $t=0$

$$\tau = 5 \text{ min}$$

$$C_a(s) = \frac{\mathcal{L}\{2\} + (5 \times 3)}{5s + 1} = \frac{2/s + 15}{5s + 1} = \frac{2}{s(5s + 1)} + \frac{15}{5s + 1}$$

For Energy Balance

$$\tau \frac{dT}{dt} + T = T_3 + \frac{Q}{\rho V_3 C_p} \quad \text{--- Energy Balance}$$

$$\tau [sT(s) - T(0)] + T(s) = T_3(s) + \frac{Q(s)}{\rho V_3 C_p}$$

$$(\tau s + 1)T(s) - \tau T(0) = T_3(s) + \frac{Q(s)}{\rho V_3 C_p}$$

$$T(s) = \frac{T_3(s) + Q(s)/\rho V_3 C_p + \tau T(0)}{\tau s + 1} = \frac{\frac{35}{s} + \frac{1.05 \times 10^6}{s(1000 \times 30 \times 1)} + 5 \times 80}{5s + 1}$$

After disturbance $T_3 = 35^\circ\text{C} \Rightarrow T(s) = 35/s$

$$T(0) = 80^\circ\text{C}$$

$$Q(s) = (1.05 \times 10^6)/s$$

$$\tau = 5$$

$$T(s) = \frac{T_3(s) + Q(s)/sV_3C_p + \tau T(0)}{\tau s + 1} = \frac{\frac{35}{s} + \frac{1.05 \times 10^6}{s(1000 \times 30 \times 1)} + 5 \times 70}{5s + 1}$$

$$T(s) = \frac{\frac{70}{s} + 400}{5s + 1} = \frac{70}{s(5s + 1)} + \frac{400}{5s + 1}$$

Solve for $\frac{dn}{dt} + 3n = 0$

with $n(0) = 2$

1) Take Laplace for Both sides
incorporating initial
conditions

2) Solve algebraically for
unknown function

$$\frac{dx}{dt} + 3x = 0$$

$$x(0) = 2$$

$$1) \quad \mathcal{L}\left\{\frac{dx}{dt}\right\} + \mathcal{L}\{3x\} = \mathcal{L}\{0\}$$

$$\Rightarrow [s x(s) - x(0)] + 3x(s) = 0$$

$$s x(s) - 2 + 3x(s) = 0$$

$$2) \quad x(s) = \frac{2}{s+3}$$

$$3) \quad \text{Invert to time domain} \quad x(t) = 2e^{-3t} \quad \text{using} \quad e^{-at} u(t) = \frac{1}{s+a}$$

Find $x(s)$ for the following differential eqn

$$\frac{d^2 x}{dt^2} + 4 \frac{dx}{dt} + 3x = u(t)$$

$$\text{with } x(0) = x'(0) = 0$$